

Voice-Based Examination System

Updated Detailed Development Roadmap

This document defines the updated engineering roadmap for the Voice-Based Examination System project. It reflects the current implementation progress, completed architecture improvements, Whisper API integration, pending production-level systems, newly added workflow features, and future deployment goals.

PHASE 0 — Architecture Cleanup

Status: ■ *COMPLETED*

- Refactored project into modular architecture
 - Implemented Flask Blueprint architecture
 - Split monolithic voice.js into modular voice engine
 - Separated routes, services, middleware, validators, utils, and database layer
 - Organized static/js and static/css structure
 - Added middleware layer
 - Added service-oriented backend architecture
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PHASE 1 — Core Exam Engine Stabilization

Status: ■ *MOSTLY COMPLETED*

Completed:

- Autosave engine
- Answer versioning
- Save-status indicators
- Submission APIs
- Session validation
- Encrypted answer storage
- Integrity verification

New Features:

- Evaluation locking system
- Re-evaluation controls
- Assignment lifecycle tracking
- Completed assignment dashboards

Pending:

- Offline save queue
 - Advanced crash recovery
 - Transaction-safe submissions
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PHASE 2 — Voice Engine Stabilization

Status: ■ *MOSTLY COMPLETED (~75–80%)*

Completed:

- recognition.js
- transcript.js
- normalization.js
- command_matcher.js
- autosave.js
- recovery.js
- tts.js
- anti_cheat.js

Pending:

- Command state machine
- Confidence thresholds
- Command cooldowns
- Fuzzy matching
- Destructive confirmations
- Multilingual preparation

New Feature:

- Previous question voice command
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PHASE 3 — Whisper Integration & Advanced Speech Layer

Status: ■ *PARTIALLY COMPLETED*

Completed:

- /api/transcribe
- Audio upload handling
- Whisper API integration

Pending:

- Streaming transcription
 - Noise filtering
 - Voice activity detection
 - Real-time streaming
 - Indian English optimization
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PHASE 4 — Academic System Features

Status: ■ *MOSTLY PENDING*

- Semester-wise exams
 - Subject collections
 - Academic analytics
 - SGPA/CGPA systems
 - PDF result generation
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PHASE 5 — Security & Audit Systems

Status: ■ *PARTIALLY COMPLETED*

Completed:

- Encrypted answer storage
- Session validation
- Role middleware

Pending:

- Audit logging
 - CSRF protection
 - OTP systems
 - Activity monitoring
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PHASE 6 — AI & Anti-Cheat Features

Status: ■ *EARLY IMPLEMENTATION*

Completed:

- AI grading base
- Anti-cheat structure

Pending:

- Rubric-based grading
 - Webcam monitoring
 - Suspicious noise detection
 - Live invigilator dashboard
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PHASE 7 — Production Readiness

Status: ■ *NOT STARTED*

- Docker setup
- docker-compose
- CI/CD pipeline

- Integration testing
- Deployment documentation

Updated Priority Order

1. Command state machine
2. Evaluation locking
3. Previous question command
4. Fuzzy command matching
5. Command confirmation
6. Assignment lifecycle
7. Confidence thresholds
8. Audit logging
9. WebSocket monitoring
10. Docker deployment

Current Project Maturity

- Architecture → Advanced
- Voice System → Strong
- Reliability → Intermediate-Advanced
- Security → Intermediate
- AI Integration → Intermediate
- Academic Features → Early
- Production Readiness → Early
- Scalability → Intermediate